

LOW STYRENE SBR PATENT LANDSCAPE: SOLUTION, EMULSION, AND FUNCTIONALIZED COPOLYMER ROUTES

1. ABSTRACT

This report synthesizes nine primary patents relevant to low styrene styrene-butadiene rubber (SBR) and closely related styrene-butadiene copolymers. The target is defined as SBR materials with styrene content below conventional grades, emphasizing reduced styrene incorporation, lower aromatic monomer fraction, and controlled monomer feed ratios. Across the landscape, the patents span both solution polymerization and emulsion polymerization, with several also addressing downstream compounding, oil replacement, filler interaction, and post-polymerization functionalization. A recurring theme is that low-styrene SBR is not only a composition target but also a process target: styrene content, sequence distribution, molecular weight, and microstructure are tuned to balance rolling resistance, wet grip, wear resistance, and processability.

The solution-polymerization patents (US11591428B2, EP1274737B1, EP2778184A1, US10174133B2, US6984706B2, US7612144B2) disclose hydrocarbon-solvent systems using organolithium initiators, polar modifiers, and in several cases controlled monomer addition, branching, or terminal functionalization. These routes generally provide tighter control over styrene content and vinyl content, with disclosed styrene levels ranging from about 20 to 40 wt% in many embodiments and specific examples near 20–26 wt% for low-styrene embodiments. The emulsion patent US8530017B2 uses water-based radical emulsion polymerization with a reactive emulsifier, showing that low-styrene SBR can also be produced in aqueous media while introducing polar functionality. US6365668B1 focuses on post-polymerization carboxylation of solution SBR and demonstrates how bound styrene content and filler compatibility can be combined with downstream water/steam stripping and silica masterbatch formation. US20060167160A1 is primarily a formulation patent, but it includes a high-styrene SBR reference and a low-styrene-relevant liquid SBR blend component, illustrating how styrene level and resin/oil selection affect tire performance.

Overall, the evidence supports low-styrene SBR synthesis via solution polymerization as the most controllable route, with emulsion polymerization remaining important where aqueous processing and reactive emulsifiers are desired. Recommended synthesis parameters from the evidence include hydrocarbon solvent solution polymerization, organolithium initiation, controlled styrene feed in the low-to-mid 20 wt% range where disclosed, and downstream coagulation/steam stripping or solvent removal as appropriate.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US11591428B2
- Patent Title: Synthesis of low molecular weight tapered styrene-butadiene copolymer and its ...
- Assignee: Bridgestone Corporation
- Jurisdiction: US
- Publication Year: 2023
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses solution polymerization of styrene and 1,3-butadiene in hexane using n-butyl lithium and sodium dodecylbenzenesulfonate, followed by coagulation in isopropanol and drum drying. The examples prepare tapered styrene-butadiene copolymers with controlled styrene sequence distribution and low molecular weight for use as oil replacements in rubber compounds. - Hexane: 0.18 kg — Example 1 - Styrene: 0.36 kg of 31.8 wt % in hexane — Example 1 - 1,3-Butadiene: 1.68 kg of 20.2 wt % in hexane — Example 1 - SDBS: 2.27 mL of 0.5 M in ethylbenzene — Example 1 - n-Butyl lithium: 6.05 mL of 1.60 M in hexane — Example 1 - Reaction temperature: 60° C. — Example 1 - Peak batch temperature: 69° C. — Example 1 - Polymerization time: 24 minutes + 2 hours — Example 1 - Styrene content: 26.6% — Example 1 - Mn: 52 kg/mol — Example 1 - Mw: 55 kg/mol — Example 1 - PDI: 1.05 — Example 1 - Tg: -76.07° C. — Example 1 - Hexane: 0.30 kg — Example 2 - Styrene: 0.55 kg of 32.9 wt % in hexane — Example 2 - 1,3-Butadiene: 2.72 kg of 20 wt % in hexane — Example 2 - SDBS: 3.63 mL of 0.5 M in ethylbenzene — Example 2 - n-Butyl lithium: 9.07 mL of 1.60 M in hexane — Example 2 - Reaction temperature: 66° C. — Example 2 - Peak batch temperature: 107° C. — Example 2 - Polymerization time: 20 minutes + 40 minutes — Example 2 - Styrene content: 25.7% — Example 2 - Mn: 66 kg/mol — Example 2 - Mw: 74 kg/mol — Example 2 - PDI: 1.12 — Example 2 - Tg: -78.79° C. — Example 2 - Total styrene content: 20 to 30 percent by weight — Claim 1 - Total butadiene content: 70 to 80 percent by weight — Claim 1 - Styrene sequences >3 units: greater than 30 weight percent of total styrene content — Claim 1 - Styrene sequences >10 units: less than 20 weight percent of total styrene content — Claim 1 - Number average molecular weight: 30,000 to 80,000 — Claim 1 - Total styrene content: 22 to 27 percent by weight — Claim 2 - Glass transition temperature: -80° C. to -70° C. — Claim 5 - Medium & Water Role: Organic hydrocarbon solution polymerization in hexane; no water-related process step disclosed in the extracted evidence. - styrene content below conventional SBR grades: 20 to 30 percent by weight; examples at 26.6% and 25.7%

Medium & Water Role - Organic hydrocarbon solution polymerization in hexane; no water-related process step disclosed in the extracted evidence. - Core reaction medium: Organic hydrocarbon solvent solution polymerization (hexane).

styrene content below conventional SBR grades - 20 to 30 percent by weight; examples at 26.6% and 25.7%

Relevant Experimental Evidence - Example 1: tapered SBR with 26.6% styrene and Mn 52 kg/mol was obtained by hexane solution polymerization and coagulation in isopropanol. - Example 2: similar solution polymerization gave 25.7% styrene SBR with Mn 66 kg/mol and Tg -78.79° C. - Example 1-2: the tapered copolymers improved wear resistance when used as oil replacements in tire rubber compositions.

Technical Relevance This patent is directly relevant because it discloses low-to-moderate styrene SBR made by controlled solution polymerization and explicitly defines a low-styrene window around 20–30 wt%. The tapered sequence design is especially useful for balancing processability and tire performance while keeping styrene content below conventional higher-styrene grades.

Patent 2

- Patent Number: US8530017B2
- Patent Title: ... copolymer using reactive emulsifier and styrene-butadiene copolymer prepared by ...
- Assignee: Korea Kumho Petrochemical Co., Ltd.
- Jurisdiction: US
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses radical emulsion polymerization of styrene and 1,3-butadiene in water using a reactive emulsifier that also participates in polymerization. The process is carried out in a pressurized reactor at low temperature, followed by termination, stripping, and drying to obtain functionalized SBR suitable for tire tread use. - Water: 1600 mL — Example 1 - Potassium rosinate: 11 g — Example 1 - Fatty acid potassium salt: 18 g — Example 1 - Potassium chloride: 1.5 g — Example 1 - Styrene: 260 g — Example 1 - Methylcarboxylate methacrylate: 30 g — Example 1 - 1,3-Butadiene: 710 g — Example 1 - EDTA: 0.5 g — Example 1 - Sodium hydrosulfite: 0.3 g — Example 1 - Ferrous sulfate: 0.1 g — Example 1 - Sodium formaldehyde sulfonate: 0.4 g — Example 1 - n-Dodecyl mercaptan: 1.0 g — Example 1 - Methane hydroperoxide: 0.6 g — Example 1 - 5-L pressurized reactor: 5 L — Example 1 - Polymerization time: 8 hours — Example 1 - Temperature: 10° C. — Example 1 - Diethylhydroxyamine: 1.0 g — Example 1 - Weight average molecular weight: 860,000 g/mol — Example 1 - Water: 1600 mL — Example 2 - Potassium rosinate: 11 g —

Example 2 - Fatty acid potassium salt: 18 g — Example 2 - Potassium chloride: 1.5 g — Example 2 - Styrene: 280 g — Example 2 - Polyoxyethylene phosphate methacrylate: 10 g — Example 2 - 1,3-Butadiene: 710 g — Example 2 - EDTA: 0.5 g — Example 2 - Sodium hydrosulfite: 0.3 g — Example 2 - Ferrous sulfate: 0.1 g — Example 2 - Sodium formaldehyde sulfonate: 0.4 g — Example 2 - n-Dodecyl mercaptan: 1.0 g — Example 2 - Methane hydroperoxide: 0.6 g — Example 2 - 5-L pressurized reactor: 5 L — Example 2 - Polymerization time: 8 hours — Example 2 - Temperature: 10° C. — Example 2 - Diethylhydroxyamine: 1.0 g — Example 2 - Weight average molecular weight: 820,000 g/mol — Example 2 - Styrene monomer: 10-50 wt % based on total reactants — Claim 3 - Butadiene monomer: 45-85 wt % based on total reactants — Claim 5 - Reactive emulsifier: polyoxyethylene phosphate methacrylate / related species — Claim 6 - Polymerization temperature: 0-80° C. — Description - Polymerization time: 1-48 hours — Description - Weight average molecular weight: 100,000-3,000,000 g/mol — Description - Medium & Water Role: Water is part of the polymerization medium in the emulsion process, and later used in stripping/workup before drying. - styrene content below conventional SBR grades: 10-50 wt % based on total reactants; examples use 260 g styrene with 710 g butadiene and 280 g styrene with 710 g butadiene

Medium & Water Role - Water is part of the polymerization medium in the emulsion process, and later used in stripping/workup before drying. - Core reaction medium: Aqueous emulsion polymerization.

styrene content below conventional SBR grades - 10-50 wt % based on total reactants; examples use 260 g styrene with 710 g butadiene and 280 g styrene with 710 g butadiene

Relevant Experimental Evidence - Example 1: water-based emulsion polymerization with 260 g styrene and 710 g butadiene produced an SBR of 860,000 g/mol. - Example 2: increasing styrene to 280 g and using polyoxyethylene phosphate methacrylate gave 820,000 g/mol polymer. - Examples 1-2: reactive emulsifiers enabled functionalized SBR while maintaining emulsion stability and tire-tread relevance.

Technical Relevance This patent is relevant because it provides an aqueous emulsion route to SBR with explicit styrene feed levels in the low-to-mid range and a reactive emulsifier strategy for property control. It is useful for low-styrene SBR when emulsion processing and polar functionality are desired.

Patent 3

- Patent Number: EP1274737B1
- Patent Title: Process for the preparation of sbr rubbers with an improved processability and ...
- Assignee: Polimeri Europa S.p.A.
- Jurisdiction: EP
- Publication Year: 2008
- Legal Status: Unknown
- Polymer Type: Not disclosed

- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses anionic solution polymerization of butadiene and styrene in hydrocarbon solvent using lithium alkyl initiators, randomizing agents, and controlled addition of a lithium alkyl followed by an alkyl bromide to create branched SBR. The polymer cement is then stripped with steam/water and dried to recover oil-extended rubber with controlled styrene content and microstructure. - Cyclohexane: 15 litres (11.7 Kg) — Comparative Example 1 - THFA: 2.4 grams (200 ppm) — Comparative Example 1 - Butadiene: 1.14 Kg — Comparative Example 1 - Styrene: 0.38 Kg — Comparative Example 1 - Temperature: 30°C — Comparative Example 1 - Li n-butyl: 2.05 grams of a solution at 5% by weight — Comparative Example 1 - Reaction time: 20 minutes — Comparative Example 1 - Temperature: 80°C — Comparative Example 1 - Ethyl alcohol: 0.15 grams — Comparative Example 1 - Highly aromatic oil: 0.560 Kg — Comparative Example 1 - Water: stirred reactor containing water heated by vapour at 100°C — Comparative Example 1 - Vacuum oven: 60°C for 24 hours — Comparative Example 1 - Residual water content: less than 0.3% by weight — Comparative Example 1 - Cyclohexane: 15 litres (11.7 Kg) — Comparative Example 3 - THFA: 4.9 grams (400 ppm) — Comparative Example 3 - Butadiene: 1.14 Kg — Comparative Example 3 - Styrene: 0.38 Kg — Comparative Example 3 - Temperature: 30°C — Comparative Example 3 - Li n-butyl: 3 grams of a solution at 5% by weight — Comparative Example 3 - Reaction time: 20 minutes — Comparative Example 3 - Temperature: 80°C — Comparative Example 3 - SiCl₄: 1.59 grams of a solution at 5% in cyclohexane — Comparative Example 3 - Highly aromatic oil: 0.560 Kg — Comparative Example 3 - Water: stirred reactor containing water heated by vapour at 100°C — Comparative Example 3 - Vacuum oven: 60°C for 24 hours — Comparative Example 3 - Residual water content: less than 0.3% by weight — Comparative Example 3 - Residence time: 30 minutes — Comparative Example 2 - Operating pressure: 3 bars — Comparative Example 2 - Butadiene: 9% by weight — Comparative Example 2 - Styrene: 3% by weight — Comparative Example 2 - Randomizing agent/vinyl promoter: 250 ppm — Comparative Example 2 - Lithium/randomizing agent ratio: 0.3-0.25 — Comparative Example 2 - Temperature: 57°C — Comparative Example 2 - Conversion: 74% — Comparative Example 2 - Temperature: 66°C — Comparative Example 2 - Overall conversion: 99% — Comparative Example 2 - Highly aromatic oil: 27% — Comparative Example 2 - Mw: 540,000 — Comparative Example 2 - Dispersion: 1.9 — Comparative Example 2 - Vinyl unit content: 67% — Comparative Example 2 - Mooney viscosity: 51 — Comparative Example 2 - Residence time: 30 minutes — Comparative Example 4 - Butadiene: 9% by weight — Comparative Example 4 - Styrene: 3% by weight — Comparative Example 4 - Randomizing agent/vinyl promoter: 350 ppm — Comparative Example 4 - Lithium/randomizing agent ratio: 0.5-0.6 — Comparative Example 4 - Temperature: 53°C — Comparative Example 4 - Conversion: 59% — Comparative Example 4 - Temperature: 61°C — Comparative Example 4 - Overall conversion: 90% — Comparative Example 4 - Temperature: 66°C — Comparative Example 4 - Final conversion: 98% — Comparative Example 4 - SiCl₄: 40% of lithium fed (equivalents/equivalents) — Comparative Example 4 - Highly aromatic oil: 27% — Comparative Example 4 - Mw: 720,000 — Comparative Example 4 - Dispersion: 2.6 — Comparative Example 4 - Vinyl unit content: 66% — Comparative Example 4 - Mooney viscosity: 50 — Comparative Example 4 - Residence time: 30' each — Comparative Example 9 - Butadiene: 9% by weight — Comparative Example 9 - Styrene: 3% by weight — Comparative Example 9 -

Randomizing agent/vinyl promoter: 350 ppm — Comparative Example 9 - Lithium/randomizing agent ratio: 1.8-2.3 — Comparative Example 9 - DVB ratio first/second reactor: 1.4-1.5 — Comparative Example 9 - Li/DVB ratio first reactor: 6.0 to 3.0 — Comparative Example 9 - Temperature: 55-57°C — Comparative Example 9 - Conversion: 62-64% — Comparative Example 9 - Temperature: 64-66°C — Comparative Example 9 - Overall conversion: 98% — Comparative Example 9 - Temperature: 67°C — Comparative Example 9 - Final conversion: 99.9% — Comparative Example 9 - Highly aromatic oil: 27.3% — Comparative Example 9 - Mw: 650,000 — Comparative Example 9 - Dispersion: 2.3 — Comparative Example 9 - Vinyl unit content: 66% — Comparative Example 9 - Mooney viscosity: 51 — Comparative Example 9 - Styrene content: 25% by weight — Table 1A - Vinyl units: 65-67% — Table 1A - Oil content: 27.3% by weight — Table 1A - Medium & Water Role: Polymerization occurs in cyclohexane solution; water is used later in steam stripping to remove solvent and recover the rubber. - styrene content below conventional SBR grades: 3% by weight in continuous feed; 25% by weight in the characterized materials

Medium & Water Role - Polymerization occurs in cyclohexane solution; water is used later in steam stripping to remove solvent and recover the rubber. - Core reaction medium: Organic hydrocarbon solvent solution polymerization (cyclohexane).

styrene content below conventional SBR grades - 3% by weight in continuous feed; 25% by weight in the characterized materials

Relevant Experimental Evidence - Comparative Example 1: batch solution SBR with 0.38 kg styrene and 1.14 kg butadiene was prepared and steam-stripped from water. - Comparative Example 2: continuous solution polymerization used 9 wt% butadiene and 3 wt% styrene feed, giving 67% vinyl and 99% conversion. - Comparative Example 4: radial continuous polymerization with SiCl₄ coupling gave 720,000 Mw and 66% vinyl. - Example 5-8: branched batch/continuous routes used lithium alkyl plus alkyl bromide to create branched SBR with 25% styrene and 64-67% vinyl.

Technical Relevance This patent is relevant because it shows low-styrene solution SBR synthesis in hydrocarbon solvent with styrene feed as low as 3 wt% in continuous operation and 25 wt% in the characterized materials. The branching strategy and solvent stripping are important process tools for tuning processability without increasing styrene content.

Patent 4

- Patent Number: EP2778184A1
- Patent Title: Solution polymerization prepared styrene/butadiene elastomer containing liquid ...
- Assignee: The Goodyear Tire & Rubber Company
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses anionically initiated organic solution polymerization of styrene and 1,3-butadiene in hexane, followed by addition of a liquid styrene/butadiene polymer or petroleum oil extension and solvent removal by steam stripping. The examples focus on high molecular weight SSBR cement extended with liquid polymer, but the base polymerization route is a controlled solution process relevant to low-styrene SBR design. - Reactor volume: 2000 liter — Example I - Temperature: 60°C to 70°C — Example I - Internal pressure: 350 kPa to 450 kPa — Example I - Monomer solution: 15 weight percent total monomers — Example I - Monomer composition: 60 weight percent 1,3-butadiene and 40 weight percent styrene — Example I - TMEDA: 0.12 pphm — Example I - n-BuLi: 0.039 pphm — Example I - Conversion: 90 to 99 percent — Example I - LSBP: 5 to 60 parts by weight per 100 parts by weight H-SSBR — Claim 1 - H-SSBR Mn: 200,000 to 1,000,000 grams/mole — Claim 1 - LSBP Mn: 1,000 to 50,000 grams/mole — Claim 1 - S + V/2: 45 to 65 or 50 to 60 — Claim 3 - Temperature: 20° to 100°C — Description - Temperature: 70°C — Description - Temperature: 100° C. to 170° C. — Desolventization description - Hexane solution: 15 weight percent total monomers — Example I - Bound styrene content: 38.9 weight percent — Example I - Vinyl content: 27.6 percent — Example I - Mooney viscosity ML(1+4), 23°C: 121 — Example I - Number average molecular weight: 350,000 — Example I - LSBP extension: 36.8 pphr — Example I - Steam stripping: 400 liter stripper — Example I - Medium & Water Role: The polymerization medium is organic solvent (hexane); water appears later in steam stripping to remove solvent from the cement before drying. - styrene content below conventional SBR grades: Not disclosed in extracted evidence

Medium & Water Role - The polymerization medium is organic solvent (hexane); water appears later in steam stripping to remove solvent from the cement before drying. - Core reaction medium: Organic solvent solution polymerization in hexane.

styrene content below conventional SBR grades - Not disclosed in extracted evidence

Relevant Experimental Evidence - Example I: a 60/40 butadiene/styrene solution polymerization in hexane produced H-SSBR with 38.9 wt% bound styrene and Mn 350,000. - Example I: the H-SSBR cement was extended with 36.8 pphr petroleum oil and recovered by steam stripping and expeller drying. - Claims: the patent defines H-SSBR and LSBP molecular-weight windows and an S+V/2 range for the liquid polymer.

Technical Relevance This patent is relevant because it establishes a solution-polymerized SSBR platform with explicit control over styrene fraction, molecular weight, and cement handling. Although the examples are not low-styrene, the disclosed solution route and compositional control are directly applicable to lower-styrene SBR design.

Patent 5

- Patent Number: US10174133B2
- Patent Title: Modified styrene-butadiene copolymer, preparation method thereof, and rubber ...
- Assignee: Lg Chem, Ltd.

- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses solution polymerization of styrene and 1,3-butadiene in normal hexane with a polar additive, followed by terminal capping, modifier addition, ethanol termination, and water-assisted solvent removal. The resulting modified styrene-butadiene copolymers are designed for improved filler interaction and combination efficiency. - Styrene: 29.7 g — Example 1 - 1,3-Butadiene: 78.1 g — Example 1 - Normal hexane: 550 g — Example 1 - 2,2-di(2-tetrahydrofuryl)propane (DTP): 0.124 g — Example 1 - Temperature: 50° C. — Example 1 - n-Butyllithium: 3 mmol — Example 1 - Additional 1,3-butadiene: 2.2 g — Example 1 - Modifier: 0.181 g — Example 1 - [DTP]/[act. Li]: 2.3 — Example 1 - [modifier]/[act. Li]: 1.0 — Example 1 - Styrene: 270 g — Example 2 - 1,3-Butadiene: 710 g — Example 2 - Normal hexane: 5 kg — Example 2 - 2,2-di(2-tetrahydrofuryl)propane (DTP): 0.94 g — Example 2 - Temperature: 40° C. — Example 2 - n-Butyllithium: 25.40 g (2.62 wt % in hexane, 33% activation) — Example 2 - Additional 1,3-butadiene: 20.0 g — Example 2 - Modifier: 1.98 g — Example 2 - [DTP]/[act. Li]: 1.46 — Example 2 - [modifier]/[act. Li]: 0.85 — Example 2 - Water: warm water heated using steam — Example 1 - Water: warm water heated using steam — Example 2 - Weight average molecular weight: 120×10^4 — Table 1, Example 1 - Number average molecular weight: 49×10^4 — Table 1, Example 1 - Weight average molecular weight: 42×10^4 — Table 1, Example 2 - Number average molecular weight: 27×10^4 — Table 1, Example 2 - Combination efficiency: 77% — Table 1, Example 1 - Combination efficiency: 64% — Table 1, Example 2 - Combination efficiency: 53% — Table 1, Comparative Example - Molecular weight distribution: 2.4 — Table 1, Example 1 - Molecular weight distribution: 1.4 — Table 1, Example 2 - Molecular weight distribution: 1.3 — Table 1, Comparative Example - Medium & Water Role: The polymerization is carried out in hexane solution; water is used later as warm steam-heated water to remove solvent before roll drying. - styrene content below conventional SBR grades: Not disclosed in extracted evidence

Medium & Water Role - The polymerization is carried out in hexane solution; water is used later as warm steam-heated water to remove solvent before roll drying. - Core reaction medium: Organic solvent solution polymerization in normal hexane.

styrene content below conventional SBR grades - Not disclosed in extracted evidence

Relevant Experimental Evidence - Example 1: low-monomer solution polymerization with DTP and a silane modifier produced a modified SBR with high combination efficiency. - Example 2: higher-scale solution polymerization with DTP and a different modifier gave a modified SBR with 64% combination efficiency. - Comparative Example: conventional modifier gave lower combination efficiency than the inventive modifiers.

Technical Relevance This patent is relevant because it discloses a controllable solution-polymerization route for modified SBR with explicit monomer charges and polar additive ratios. The process is useful for low-styrene SBR development because styrene and butadiene feed levels can be directly adjusted in a hydrocarbon solvent system.

Patent 6

- Patent Number: US6984706B2
- Patent Title: Diene rubber, process for production thereof, rubber compositions, process for ...
- Assignee: Zeon Corporation
- Jurisdiction: US
- Publication Year: 2006
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses living anionic solution polymerization of diene monomers, including styrene-butadiene copolymers, in hydrocarbon solvent with controlled polymerization stopping to tailor molecular weight distribution and processability. Water is used in some examples to deactivate active terminals in the monomer feed and in steam stripping of the polymer solution, but the core synthesis remains solution polymerization. - Cyclohexane: 4,000 parts — Example 1 - Styrene: 190 parts — Example 1 - 1,3-Butadiene: 510 parts — Example 1 - Tetramethylethylenediamine: 2.0 parts — Example 1 - n-Butyllithium: 0.092 part — Example 1 - Polymerization temperature: 45° C. — Example 1 - Conversion: 23% at 30 minutes — Example 1 - Additional monomer mixture: 60 parts styrene and 240 parts 1,3-butadiene — Example 1 - Water deactivation: 25% by mole of active terminal — Example 1 - Tetramethoxysilane: 0.013 part — Example 1 - Methanol: 0.09 part — Example 1 - Highest temperature: 65° C. — Example 1 - Styrene content: 25% — Table 1 - Vinyl bond content: 60% — Table 1 - Mw: 85.5 ×1000 — Table 1 - Mw/Mn: 1.69 — Table 1 - Cyclohexane: 4,000 parts — Example 4 - Styrene: 320 parts — Example 4 - 1,3-Butadiene: 380 parts — Example 4 - Tetramethylethylenediamine: 0.14 parts — Example 4 - n-Butyllithium: 0.092 part — Example 4 - Water deactivation: 25% by mole of active terminal — Example 4 - Tetramethoxysilane: 0.020 part — Example 4 - Highest temperature: 65° C. — Example 4 - Styrene content: 35% — Table 1 - Vinyl bond content: 40% — Table 1 - Mw: 86.5 ×1000 — Table 1 - Mw/Mn: 1.78 — Table 1 - Cyclohexane: 1.12 kg — Example 2/3 style production examples not fully extracted in evidence - Styrene content: 21.7% — Example 2/3 style data in Table 1 - Styrene content: 22.2% — Example 3 style data in Table 1 - Vinyl content: 32.1% — Table 1 - Vinyl content: 31.2% — Table 1 - Mooney viscosity: 49-51 — Table 1 - Water: added to monomer mixture to deactivate 7% or 8% by mole of active terminal — Comparative Examples 2 and 4 - Steam stripping: solvent removed by stripping in a vapour stream — Comparative Examples 1, 3, 4 - Medium & Water Role: The core polymerization is in cyclohexane solution; water is used both to deactivate active terminals in the monomer feed and later in steam stripping to remove solvent. - styrene content below conventional SBR grades: 25% styrene; 21.7% styrene; 22.2% styrene

Medium & Water Role - The core polymerization is in cyclohexane solution; water is used both to deactivate active terminals in the monomer feed and later in steam stripping to remove solvent. - Core reaction medium: Organic hydrocarbon solvent solution polymerization (cyclohexane).

styrene content below conventional SBR grades - 25% styrene; 21.7% styrene; 22.2% styrene

Relevant Experimental Evidence - Example 1: batch solution SBR with 25% styrene and 60% vinyl bond content was prepared using controlled monomer addition and water deactivation. - Example 4: a higher-styrene batch example gave 35% styrene and 40% vinyl bond content. - Table 1: the patent compares linear, radial, and branched SBR structures with styrene contents around 25% and varying molecular weights and Mooney viscosities.

Technical Relevance This patent is relevant because it provides a controlled solution-polymerization framework for SBR with explicit low-styrene examples around 21.7–25% styrene. The process emphasizes living polymerization, controlled termination, and solvent removal, all important for tailoring low-styrene SBR properties.

Patent 7

- Patent Number: US6365668B1
- Patent Title: Rubber compounds containing solution rubbers which contain carboxyl groups
- Assignee: Bayer Aktiengesellschaft
- Jurisdiction: US
- Publication Year: 2002
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses post-polymerization carboxylation of solution SBR in cyclohexane using 3-mercaptopropionic acid and dilauroyl peroxide, followed by steam distillation to remove solvent and vacuum drying. The resulting carboxylated solution SBR is then used in rubber compounds and silica masterbatches, with water/steam used in workup and stripping rather than as the polymerization medium. - Bound styrene content: 20% by weight — Example 1 - Cyclohexane: 275 kg — Example 1 - Temperature: 80° C. — Example 1 - 3-mercaptopropionic acid: 0.563 kg — Example 1 - Dilauroylperoxide: 0.045 kg — Example 1 - Stirring time: 2 hours — Example 1 - Aromatic oil Mobilsol K: 17.12 kg — Example 1 - Steam distillation: solvent removed — Example 1 - Drying: 70° C. under vacuum — Example 1 - Mineraloil: 37.5 phr — Example 1 - Carboxyl content: 0.5 wt. % — Example 1 - Bound styrene content: 25% by weight — Example 2 - 1,2-bonded butadiene: 50% by weight — Example 2 - Cyclohexane: 4 liters — Example 2 - Temperature: 80° C. — Example 2 - 3-mercaptopropionic acid: 6.25 g — Example 2 - Dilauroyl peroxide: 0.25 g — Example 2 - Stirring time: 5 hours — Example 2 - Drying: 70° C. under vacuum — Example 2 - Mooney viscosity ML 1+4 (100° C.): 65 — Example 2 - Carboxyl content: 0.5% by weight — Example 2 - 3-mercaptopropionic acid: 3.2 g — Example 3 - 3-mercaptopropionic acid: 12.5 g — Example 4 - 3-mercaptopropionic acid: 25 g

— Example 5 - 2-mercaptopropionic acid: 25 g — Example 6 - 2-mercaptopropanoic acid: 12.5 g — Example 7 - Ammonia in water: 6.94 g of 26 wt. % — Example 8 - Stirring time: 10 min — Example 8 - Carboxyl content: 0.5 wt. % in ammonium salt form — Example 8 - Bound styrene content: 23.5% weight — Comparative Example 9 - Water: 6.94 g of 26 wt. % ammonia in water — Example 8 - Water: steam distillation — Example 10 - Water waste: free from hydrated silica — Example 10 - Medium & Water Role: The core chemistry is solution-based in cyclohexane; water is used later for steam distillation, ammonia neutralization in water, and silica masterbatch workup. - styrene content below conventional SBR grades: 20% by weight; 25% by weight; 23.5% weight

Medium & Water Role - The core chemistry is solution-based in cyclohexane; water is used later for steam distillation, ammonia neutralization in water, and silica masterbatch workup. - Core reaction medium: Organic solvent solution polymerization / post-polymerization modification in cyclohexane.

styrene content below conventional SBR grades - 20% by weight; 25% by weight; 23.5% weight

Relevant Experimental Evidence - Example 1: 20 wt% bound styrene solution SBR was carboxylated and oil-extended, then steam-stripped and vacuum dried. - Example 2: 25 wt% bound styrene solution SBR was carboxylated to 0.5 wt% carboxyl content and obtained with Mooney 65. - Example 8: carboxylated solution SBR was neutralized with ammonia in water to form ammonium carboxylate groups. - Example 10: carboxylated solution SBR was combined with precipitated silica and aromatic oil to form a masterbatch, then steam-stripped. - Comparative Example 9: emulsion-SBR was carboxylated for comparison and showed higher Mooney viscosity than solution SBR.

Technical Relevance This patent is relevant because it includes solution SBR starting materials with bound styrene contents of 20–25 wt%, squarely within low-styrene territory, and shows how post-functionalization and silica masterbatching can be combined with such rubbers. It also demonstrates downstream water/steam handling important for industrial recovery and compounding.

Patent 8

- Patent Number: US20060167160A1
- Patent Title: Rubber composition for tire and tire made therefrom
- Assignee: Bridgestone Corporation
- Jurisdiction: US
- Publication Year: 2006
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses tire rubber compositions built from SBR, hydrogenated SBR, and tackifying resins or liquid polymers, rather than a primary SBR synthesis route. The examples show how styrene-butadiene copolymers with specified styrene contents are compounded with fillers, oils, and resins to improve road grip, hysteresis, and processability. - Copolymer (a): weight-average molecular weight 1.0×10^6 — Example 1 - Copolymer (a): bound styrene 30% by mass — Example 1 - Copolymer (a): vinyl unit in butadiene portion 50% — Example 1 - Copolymer (b): weight-average molecular weight 1.0×10^4 — Example 1 - Copolymer (b): bound styrene 40% by mass — Example 1 - Copolymer (b): hydrogenated double bond fraction 90% — Example 1 - Copolymer (C): 40 parts by mass — Example 1 - Carbon black A: 80 parts by mass — Example 1 - Styrene-butadiene copolymer rubber (a): styrene units content 23.5% — Comparative Example 23 - SBR #1500 obtained by emulsion polymerization — Comparative Example 23 - Weight-average molecular weight of copolymer (a): 4.0×10^5 to 3.0×10^6 — Claim 1 - Bound styrene St(a): 10 to 50% by mass — Claim 1 - Vinyl unit in butadiene portion: 20 to 70% — Claim 1 - Weight-average molecular weight of copolymer (b): 5.0×10^3 to 2.0×10^5 — Claim 1 - Bound styrene St(b): 25 to 70% by mass — Claim 1 - Hydrogenated double bond fraction: 60% or greater — Claim 1 - Liquid polymer molecular weight: 1,000 to 50,000 — Claim 1 - St(b) ■ St(a)+10 — Claim 1 - St(b): 30 to 60% by mass — Claim 2 - Hydrogenated double bond fraction: 80% or greater — Claim 3 - Component (B): 10 to 200 parts by mass — Claim 1 - Component (C): 1,000 to 50,000 molecular weight — Claim 1 - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - styrene content below conventional SBR grades: 30% by mass; 23.5% styrene in Comparative Example 23

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: Not a synthesis patent; formulation/compounding of rubber compositions.

styrene content below conventional SBR grades - 30% by mass; 23.5% styrene in Comparative Example 23

Relevant Experimental Evidence - Example 1: a 30% styrene SBR was compounded with liquid hydrogenated SBR and petroleum resin to improve tire grip and hysteresis. - Examples 2-9: alternative tackifying resins were used with the same SBR/hydrogenated SBR framework. - Comparative Example 23: emulsion SBR #1500 was used as a reference material with 23.5% styrene content.

Technical Relevance This patent is relevant mainly as a formulation reference because it explicitly uses SBR with 30% styrene and cites an emulsion SBR at 23.5% styrene. It helps define the lower end of practical styrene levels in tire compounds, although it does not primarily teach SBR synthesis.

Patent 9

- Patent Number: US7612144B2
- Patent Title: Use of sulfur containing initiators for anionic polymerization of monomers
- Assignee: Bridgestone Corporation

- Jurisdiction: US
- Publication Year: 2009
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses anionic polymerization of styrene-butadiene and butadiene polymers using sulfur-containing initiators such as 2-lithio-2-methyl-1,3-dithiane and 2-lithio-2-phenyl-1,3-dithiane. The SBR examples are prepared in hexane with butadiene and styrene, then coagulated in isopropanol and drum dried; the initiator choice provides functional head groups that improve filler interaction and rolling resistance.

- Hexane: 1.12 kg — Example No. 2 - Styrene: 0.48 kg of 33 wt % styrene in hexane — Example No. 2 - Butadiene: 2.89 kg of 22.0 wt % butadiene in hexane — Example No. 2 - Cyclic oligomeric oxolanyl alkane modifier: 0.5 mL of 1.6 M — Example No. 2 - 2-lithio-2-methyl-1,3-dithiane: 22.63 mL of 0.234 M in tetrahydrofuran — Example No. 2 - Temperature: 24° C. — Example No. 2 - Jacket temperature: 54° C. — Example No. 2 - Peak batch temperature: 76.5° C. — Example No. 2 - Reaction time: 15 minutes + 25 minutes — Example No. 2 - Mn: 153 kg/mol — Example No. 2 - Mw: 167 kg/mol — Example No. 2 - Tg: -44.4° C. — Example No. 2 - Styrene: 21.7% — Example No. 2 - Block styrene: 1.3% — Example No. 2 - Vinyl: 32.1% — Example No. 2 - 1,4-butadiene incorporation: 46.2% — Example No. 2 - Hexane: 1.12 kg — Example No. 3 - Styrene: 0.48 kg of 33 wt % styrene in hexane — Example No. 3 - Butadiene: 2.89 kg of 22.0 wt % butadiene in hexane — Example No. 3 - 2-lithio-2-methyl-1,3-dithiane: 16.96 mL of 0.234 M in tetrahydrofuran — Example No. 3 - Temperature: 24° C. — Example No. 3 - Jacket temperature: 54° C. — Example No. 3 - Peak batch temperature: 75.7° C. — Example No. 3 - Reaction time: 17 minutes + 10 minutes — Example No. 3 - Mn: 208 kg/mol — Example No. 3 - Mw: 240 kg/mol — Example No. 3 - Tg: -43.8° C. — Example No. 3 - Styrene: 22.2% — Example No. 3 - Block styrene: 1.6% — Example No. 3 - Vinyl: 31.2% — Example No. 3 - 1,4-butadiene incorporation: 46.5% — Example No. 3 - Hexane: 1.07 kg — Example No. 4 - Styrene: 0.48 kg of 33 wt % styrene in hexane — Example No. 4 - Butadiene: 2.95 kg of 21.6 wt % butadiene in hexane — Example No. 4 - 2-methyl-1,3-dithiane: 8.47 mL of 0.5 M in hexane — Example No. 4 - Butyllithium: 3.42 mL of 1.55 M in hexanes — Example No. 4 - Temperature: 24° C. — Example No. 4 - Jacket temperature: 54° C. — Example No. 4 - Peak batch temperature: 68.6° C. — Example No. 4 - Reaction time: 28 minutes + 10 minutes — Example No. 4 - Mn: 135 kg/mol — Example No. 4 - Mw: 142 kg/mol — Example No. 4 - Tg: 56.6° C. — Example No. 4 - Hexane: 4.75 kg — Example 45 - Styrene: 1.25 kg 33% styrene in hexane — Example 45 - Butadiene: 7.55 kg 21.7 wt % butadiene in hexane — Example 45 - 2-methyl-1,3-dithiane: 37.1 mL of 0.5 M — Example 45 - Butyl lithium: 11.04 mL of 1.68 M — Example 45 - Temperature: 48.9° C. — Example 45 - Styrene: 20.2% — Example 45 - Block styrene: 2.2% — Example 45 - 1,2 butadiene: 44.9% — Example 45 - TEOS: 1 eq. per BuLi — Example 46 - 2-methylthio-2-thiazoline: 1 eq. per BuLi — Example 47 - Bu₃SnCl: 1 eq. per BuLi — Example 48 - Styrene: 20.6% — Example 46 - Styrene: 20.7% — Example 47 - Styrene: 21.0% — Example 48 - Block styrene: 2.0% — Example 46 - Block styrene: 1.9% — Example 47 - Block styrene: 2.0% — Example 48 - Medium & Water Role:

No water-related process step disclosed in the extracted evidence. - styrene content below conventional SBR grades: 21.7% styrene; 22.2% styrene; 20.2% styrene; 20.6-21.0% styrene

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: Organic solvent solution polymerization in hexane / tetrahydrofuran-containing initiator solution.

styrene content below conventional SBR grades - 21.7% styrene; 22.2% styrene; 20.2% styrene; 20.6-21.0% styrene

Relevant Experimental Evidence - Example No. 2: 21.7% styrene SBR was prepared using 2-lithio-2-methyl-1,3-dithiane and showed Tg -44.4° C. - Example No. 3: similar SBR gave 22.2% styrene and 31.2% vinyl content. - Example 45: a 20.2% styrene SBR was prepared with 2-methyl-1,3-dithiane initiation. - Examples 46-48: terminal functionalization after polymerization preserved low-styrene SBR compositions around 20.6-21.0% styrene. - Example 12/compound data: functionalized SBRs improved tan delta and filler interaction in tire compounds.

Technical Relevance This patent is highly relevant because it directly discloses low-styrene SBR examples around 20–22% styrene and shows how sulfur-containing initiators can be used to control head-group functionality without sacrificing low styrene content. The solution-polymerization route is well suited to low-styrene SBR synthesis and tire performance tuning.

PATENT COMPARISON TABLE

Patent	Applicant	Priority	Medium & Water Role	styrene content below conventional SBR grades	Key Findings
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US11591428B2	Bridgestone Corporation	2023	Organic hydrocarbon solution polymerization in hexane; no water-related process step disclosed in the extracted evidence.	20 to 30 percent by weight; examples at 26.6% and 25.7%	Example 1: tapered SBR with 26.6% styrene and Mn 52 kg/mol was obtained by hexane solution polymerization and coagulation
US8530017B2	Korea Kumho Petrochemical Co., Ltd.	2013	Water is part of the polymerization medium in the emulsion process, and later used in stripping/workup before drying.	10-50 wt % based on total reactants; examples use 260 g styrene with 710 g butadiene and 280 g styrene with 710 g butadiene	Example 1: water-based emulsion polymerization with 260 g styrene and 710 g butadiene produced an SBR of 860,000 g/mol.

EP1274737B1	Polimeri Europa S.p.A.	2008	Polymerization occurs in cyclohexane solution; water is used later in steam stripping to remove solvent and recover the rubber.	3% by weight in continuous feed; 25% by weight in the characterized materials	Comparative Example 1: batch solution SBR with 0.38 kg styrene and 1.14 kg butadiene was prepared and steam-stripped from
EP2778184A1	The Goodyear Tire & Rubber Company	2014	The polymerization medium is organic solvent (hexane); water appears later in steam stripping to remove solvent from the cement before drying.	Not disclosed in extracted evidence	Example I: a 60/40 butadiene/styrene solution polymerization in hexane produced H-SSBR with 38.9 wt% bound styrene and M

US10174133B2	Lg Chem, Ltd.	2019	The polymerization is carried out in hexane solution; water is used later as warm steam-heated water to remove solvent before roll drying.	Not disclosed in extracted evidence	Example 1: low-monomer solution polymerization with DTP and a silane modifier produced a modified SBR with high combinat
US6984706B2	Zeon Corporation	2006	The core polymerization is in cyclohexane solution; water is used both to deactivate active terminals in the monomer feed and later in steam stripping to remove solvent.	25% styrene; 21.7% styrene; 22.2% styrene	Example 1: batch solution SBR with 25% styrene and 60% vinyl bond content was prepared using controlled monomer addition

US6365668B1	Bayer Aktiengesellschaft	2002	The core chemistry is solution-based in cyclohexane; water is used later for steam distillation, ammonia neutralization in water, and silica masterbatch workup.	20% by weight; 25% by weight; 23.5% weight	Example 1: 20 wt% bound styrene solution SBR was carboxylated and oil-extended, then steam-stripped and vacuum dried.
US20060167160A1	Bridgestone Corporation	2006	No water-related process step disclosed in the extracted evidence.	30% by mass; 23.5% styrene in Comparative Example 23	Example 1: a 30% styrene SBR was compounded with liquid hydrogenated SBR and petroleum resin to improve tire grip and hy

US7612144B2	Bridgestone Corporation	2009	No water-related process step disclosed in the extracted evidence.	21.7% styrene; 22.2% styrene; 20.2% styrene; 20.6-21.0% styrene	Example No. 2: 21.7% styrene SBR was prepared using 2-lithio-2-methyl-1,3-dithia
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SUPPORTING / RELATED PATENTS (SECONDARY)

Supporting Patent 1

- Patent Number: US5902856A
- Patent Title: Rubber composition of two diene rubbers
- Assignee: Nippon Zeon Co., Ltd.
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: This patent is about a diene rubber blend including modified and branched SBR, not about low-styrene SBR manufacture itself.

Supporting Patent 2

- Patent Number: EP3030430B1
- Patent Title: Modified diene elastomer comprising a diene elastomer coupled by an ...
- Assignee: Compagnie Générale des Etablissements Michelin
- Jurisdiction: EP
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The invention is about a modified diene elastomer with chain-end and mid-chain functionalization, not specifically about low-styrene SBR manufacture.

Supporting Patent 3

- Patent Number: US7659337B2
- Patent Title: Elastomeric blend for air barriers comprising grafted resin components
- Assignee: Exxonmobil Chemical Patents Inc.
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)

- Technical Note: This is an elastomeric air-barrier composition with SBR only as a possible secondary rubber, so it is not primarily about the requested SBR material.

Supporting Patent 4

- Patent Number: US12247114B1
- Patent Title: Methods of preparing a composite having elastomer and filler
- Assignee: Beyond Lotus Llc
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The patent is about preparing composites with elastomer and filler, where SBR is only one selectable elastomer among many, so it is not primarily about SBR.

Supporting Patent 5

- Patent Number: US11753530B2
- Patent Title: Inter-penetrating elastomer network derived from ground tire rubber particles
- Assignee: William B. Coe
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The patent is about an inter-penetrating elastomer network derived from ground tire rubber particles, with SBR only listed as one possible elastomer in the network.

Supporting Patent 6

- Patent Number: US9273195B2
- Patent Title: Tires comprising cellulose ester/elastomer compositions
- Assignee: Eastman Chemical Company
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: This patent is about tires comprising cellulose ester/elastomer compositions, and the SBR mention is only as part of a broader elastomer system.

Supporting Patent 7

- Patent Number: EP2673329B1
- Patent Title: Fluid activatable adhesives and fluids for activating same for use with liner- ...
- Assignee: NuLabel Technologies, Inc.
- Jurisdiction: EP
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The patent is about adhesives using hydrophobic polymers, with SBR only listed as one possible hydrophobic polymer ingredient.

Supporting Patent 8

- Patent Number: US9926440B2
- Patent Title: Additives for rubber compositions
- Assignee: Arizona Chemical Company, LLC
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The patent is about additives for rubber compositions, with SBR only one of several possible rubber (A) options, so it is not primarily about low-styrene SBR.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- Solution-polymerized patents generally disclose lower and more controllable styrene levels than the emulsion patent, with low-styrene examples around 20-26 wt% in US11591428B2, US6365668B1, and US7612144B2.
- US8530017B2 uses aqueous emulsion polymerization with styrene feed at 10-50 wt% of total reactants, but the examples are centered at 260-280 g styrene with 710 g butadiene, i.e. a moderate-styrene emulsion SBR route.
- EP1274737B1 and US6984706B2 emphasize solution polymerization in hydrocarbon solvent with styrene contents around 25% in the characterized materials, while EP1274737B1 also shows continuous feeds as low as 3 wt% styrene in the reactor feed.
- US7612144B2 and US11591428B2 both demonstrate low-styrene solution SBR around 20-26% styrene, but US7612144B2 adds sulfur-containing initiator chemistry for head-group functionality, whereas US11591428B2 focuses on tapered sequence distribution and low molecular weight.
- US6365668B1 is distinctive in combining low-styrene solution SBR starting materials with post-polymerization carboxylation and silica masterbatching, showing that low styrene can be paired with filler affinity improvements.
- US20060167160A1 is primarily a compounding patent rather than a synthesis patent, but it confirms practical tire-compound use of SBR at 30% styrene and references an emulsion SBR at 23.5% styrene as a comparative material.
- Across the solution patents, hydrocarbon solvents such as hexane and cyclohexane dominate, with coagulation/steam stripping or isopropanol coagulation used for recovery; the emulsion patent instead uses water as the polymerization medium and stripping/workup medium.

4. CONCLUSION

The evidence supports low-styrene SBR synthesis primarily through hydrocarbon solution polymerization, where styrene content can be held near 20-26 wt% while maintaining control over

molecular weight, vinyl content, and sequence distribution. Emulsion polymerization remains viable for functionalized SBR, but the strongest low-styrene synthesis control appears in the solution-route patents. Recommended parameters from the extracted evidence are hydrocarbon solvent solution polymerization, organolithium initiation, styrene feed in the low-to-mid 20 wt% range where disclosed, controlled termination or branching as needed, and downstream coagulation or steam stripping for recovery. For tire applications, the patents indicate that low-styrene SBR benefits from tailored microstructure and, when needed, post-functionalization or filler-compatible modifiers to preserve processability and performance.

5. REFERENCES

- US11591428B2 | Synthesis of low molecular weight tapered styrene-butadiene copolymer and its ... | Bridgestone Corporation | US | 2023 | <https://patents.google.com/patent/US11591428B2/en>
- US8530017B2 | ... copolymer using reactive emulsifier and styrene-butadiene copolymer prepared by ... | Korea Kumho Petrochemical Co., Ltd. | US | 2013 | <https://patents.google.com/patent/US8530017B2/en>
- EP1274737B1 | Process for the preparation of sbr rubbers with an improved processability and ... | Polimeri Europa S.p.A. | EP | 2008 | <https://patents.google.com/patent/EP1274737B1/en>
- EP2778184A1 | Solution polymerization prepared styrene/butadiene elastomer containing liquid ... | The Goodyear Tire & Rubber Company | EP | 2014 | <https://patents.google.com/patent/EP2778184A1/en>
- US10174133B2 | Modified styrene-butadiene copolymer, preparation method thereof, and rubber ... | Lg Chem, Ltd. | US | 2019 | <https://patents.google.com/patent/US10174133B2/en>
- US6984706B2 | Diene rubber, process for production thereof, rubber compositions, process for ... | Zeon Corporation | US | 2006 | <https://patents.google.com/patent/US6984706B2/en>
- US6365668B1 | Rubber compounds containing solution rubbers which contain carboxyl groups | Bayer Aktiengesellschaft | US | 2002 | <https://patents.google.com/patent/US6365668B1/en>
- US20060167160A1 | Rubber composition for tire and tire made therefrom | Bridgestone Corporation | US | 2006 | <https://patents.google.com/patent/US20060167160A1/en>
- US7612144B2 | Use of sulfur containing initiators for anionic polymerization of monomers | Bridgestone Corporation | US | 2009 | <https://patents.google.com/patent/US7612144B2/en>